

Lab 02

Exercises

1. Write an assembly language program that asks the user to input a letter in uppercase. Then, the program should convert that letter to lowercase and print it, as shown below:

```
Enter an uppercase letter: R
The letter in lowercase is: r
```

2. Write an assembly language program that asks the user to input a letter in lowercase. Then, the program should convert that letter to uppercase and print it, as shown below:

```
Enter a lowercase letter: g
The letter in uppercase is: G
```

3. Write an assembly language program that asks the user to input two numbers with respective prompt messages and then prints their sum/addition, as shown below.

```
Enter the first number: 2
Enter the second number: 5
The addition is: 7
```

4. Write an assembly language program that asks the user to input two numbers with respective prompt messages and then prints their difference/subtraction, as shown below.

```
Enter the first number: 9
Enter the second number: 6
The subtraction is: 3
```

5. Write an assembly language program that asks the user to input two numbers with respective prompt messages. Then, the program should swap/exchange their values and print them, as shown below.

```
Enter the first number: 8
Enter the second number: 2
After swapping:
The first number is: 2
The second number is: 8
```

6. Write an assembly language program that displays the pyramid made of asterisks, as shown below:

```
*****
****
***
**
*
```

7. Write an assembly language program that displays the following output:

```
1
12
123
1234
12345
```

8. Write a program to (a) prompt the user, (b) read first, middle, and last initials of a person's name, and (c) display them down the left margin.

Sample Execution:

```
Enter three initials: JFK
```

```
J
F
K
```

9. Write a program to display a 4 x 10 solid box of asterisks (4 rows and 10 columns), as shown below:

```
*****
*****
*****
*****
```